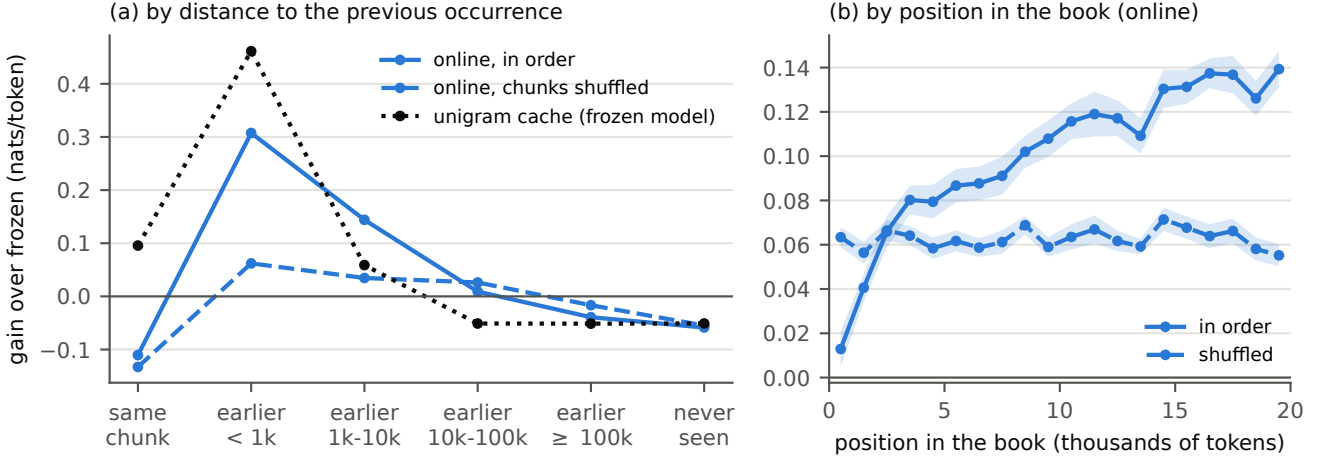
(a) by distance to the previous occurrence
gain over frozen (nats/token)
0.4
0.3
0.2
0.1
0.0
−0.1
online, in order
online, chunks shuffled
unigram cache (frozen model)
same chunk
earlier < 1k
earlier 1k-10k
earlier 10k-100k
earlier ≥ 100k
never seen
(b) by position in the book (online)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
in order
shuffled
0
5
10
15
20
position in the book (thousands of tokens)